

ANOVA

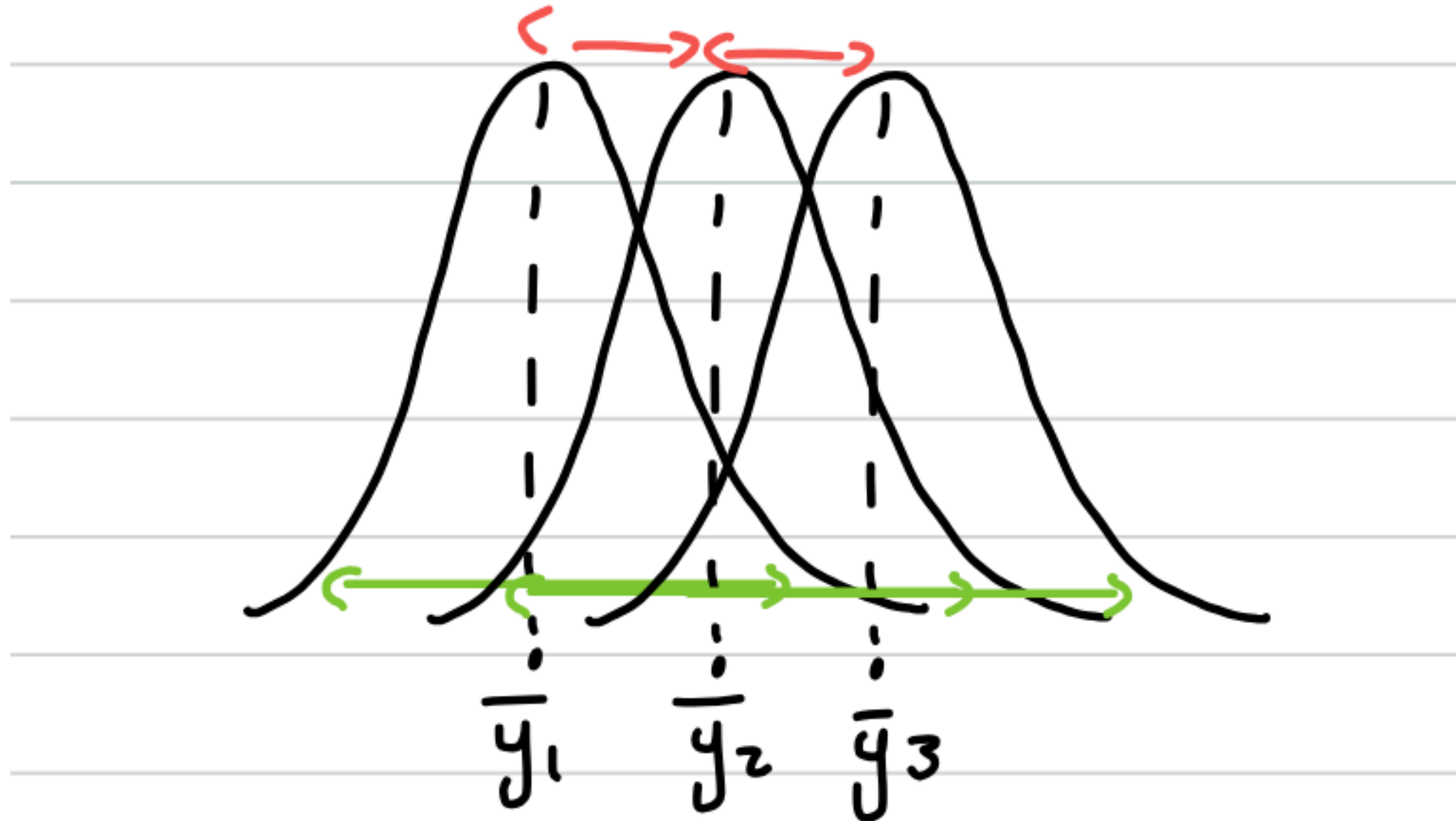
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Aim

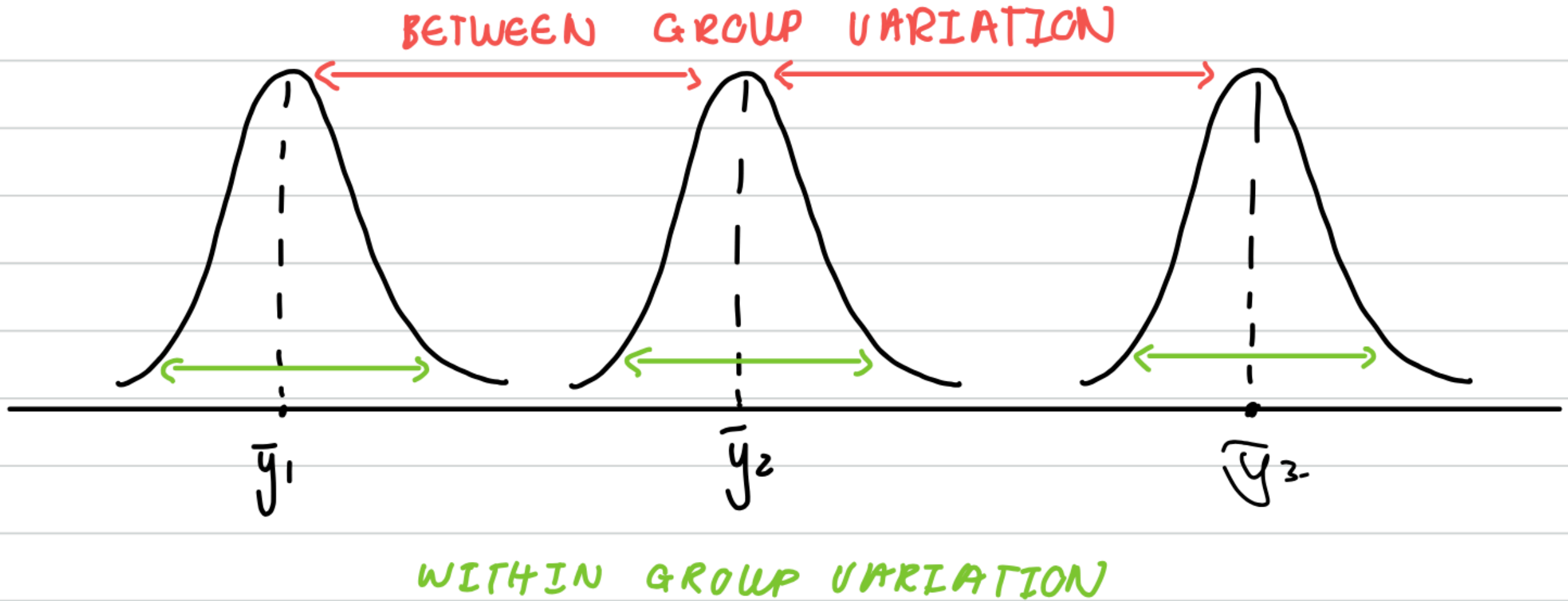
We want to know whether our deliberate manipulation of conditions (the treatments) has introduced systematic variation in the response, beyond what we would expect from random noise.

- Assume treatments had no effect: all observations come from the same population with common mean (μ)
- If treatments had an effect: observations come from populations with different means ($\mu_1, \mu_2, \dots, \mu_k$)
- Treatments “carve up” original single population into several sub-populations with different means.

Treatments had no effect



Treatments had effect



Analysis of variance

- By comparing variation between groups i.t.o differences between means
- to variation within groups
- we can make conclusions about means!

Getting to measures of between and within variability

- Use our model to decompose the total sums of square
- into sums of squares due to treatments (differences amongst means) and
- sums of squares due to error (differences within groups)